

Credit Decision Data Platform — 30 / 60 / 90-Day Execution Plan

Owner: Data Engineering Lead

Audience: Data Engineering, Platform/DevOps, Risk, Compliance, Product

Status: Plan of record for Q2 2025 go-live

Companion documents: [../SPEC.md](#) (contract) · [../architecture/architecture.md](#) (design)

1. Executive summary

We are delivering the platform specified in [SPEC.md](#): a medallion lakehouse on AWS that fuses four sources into a unified, immutable, audit-grade credit decision_input for three products (PERSONAL_FINANCE, BNPL, CARD_ALT), resolves customer identity across sources, snapshots every input immutably for UAE Central Bank audit, scores data quality, and serves a portfolio-monitoring mart to the risk team.

This plan splits work into three 30-day tranches with a hard line between **LAUNCH-CRITICAL** (everything required to go live at **10K decisions/day in Q2 2025**) and **POST-LAUNCH** (hardening, then scale to **100K/day**, EMR migration, ML scoring, advanced MDM).

- **Day 0–30 — Foundation & launch-critical.** Landing zone, medallion S3 with Object Lock, RDS + Debezium CDC, MSK, all four ingestion paths, Bronze→Silver Glue jobs, deterministic identity resolution (v1), decision_input, immutable snapshots, must-pass DQ, Athena, a minimal risk mart, Airflow orchestration. **Outcome: a thin but complete, compliant end-to-end slice that can go live.**
- **Day 31–60 — Hardening.** Probabilistic matching, full DQ scorecard + SNS alerting, portfolio mart + Metabase dashboards, backfill/replay, observability, DR, and a formal security + UAE-CB audit-trail review. **Outcome: production-grade reliability and a passed compliance review.**
- **Day 61–90 — Scale & optimise.** 100K/day load test, EMR migration evaluation, cost optimisation, partition/compaction tuning (OPTIMIZE/Z-ORDER), advanced MDM + stewardship queue, ML feature groundwork, and the post-launch roadmap. **Outcome: proven headroom to 100K/day and a costed roadmap.**

Goals

#	Goal	Tied to
G1	Go live at 10K decisions/day in Q2 2025 with all four sources fused	SPEC §1, §2
G2	Every decision immutably snapshotted, 7-yr WORM, tamper-evident	SPEC §7

#	Goal	Tied to
G3	No source data silently dropped (unresolved records materialised)	SPEC §6
G4	Must-pass DQ enforced before any row reaches the risk mart	SPEC §8
G5	Proven, costed path to 100K/day within 12 months	SPEC §1, §3
G6	Pass UAE Central Bank audit-trail + data-residency review	SPEC §7, §10

Success metrics (measured continuously post-launch)

Metric	Launch target	Scale target
Decisions processed/day	10,000	100,000
End-to-end freshness (application → decision_input)	≤ 60 min	≤ 30 min
Snapshot coverage (decisions with an Object-Lock snapshot)	100%	100%
Must-pass DQ enforcement (bad rows quarantined, not served)	100%	100%
Identity resolution rate (attached, not UNRESOLVED)	≥ 97%	≥ 99%
needs_manual_review backlog worked within SLA	≤ 2 business days	≤ 1 business day
Source-freshness SLA adherence (AECB<24h, fraud<1h, AML<6h, CDC<15m)	≥ 99%	≥ 99.9%
Pipeline cost per 1K decisions	baseline established	≥ 40% below launch baseline

2. Day 0–30 — Foundation & launch-critical

Theme: stand up a thin, complete, compliant end-to-end slice. Everything in this tranche is

LAUNCH-CRITICAL — no item here is optional for go-live.

ID	Workstream	Deliverable	SPEC ref	Launch-critical
F1	Landing zone	AWS Organizations, dev/pre/prod accounts, me-central-1 region pinned, IAM baseline, KMS keys, VPC + private subnets + NAT, CloudTrail on	§10, compliance	■
F2	Medallion S3	Bucket wio-credit-decision- <code>\${ENV}</code> with bronze/silver/gold/_checkpoints/_athena_results; SSE-KMS; lifecycle stubs	§3	■

ID	Workstream	Deliverable	SPEC ref	Launch-critical
F3	Object Lock bucket	Separate snapshot bucket, Object Lock compliance mode , default 7-yr retention, versioning	§7	■
F4	Glue catalog	Databases credit_bronze, credit_silver, credit_gold	§4	■
F5	RDS + CDC	RDS PostgreSQL (identity spine), logical replication on, Debezium source connector	§2, §6	■
F6	MSK + sink	MSK cluster, topic cdc . public.customer_profile, MSK Connect S3 sink → Delta bronze/customer_profile /	§2, §3	■
F7	Ingestion — AECB	Glue batch job: SFTP pull → parse XML → Parquet bronze/aecb/ ingest_date=...; job bookmark	§2, §3	■
F8	Ingestion — Fraud	Scheduled Glue poll job: REST GET → JSON → Parquet bronze/fraud/...; bookmark	§2	■
F9	Ingestion — AML	API Gateway + Lambda webhook receiver → Parquet bronze/aml/...	§2	■
F10	Bronze→Silver	Glue PySpark jobs → aecb_credit_report, fraud_score, aml_screening, customer_profile (typed, deduped, PII-tagged , audit block); CDC via Delta MERGE	§3, §5, §10	■
F11	Identity resolution v1	Deterministic-only : spine seeding UUIDv5(internal_customer_uuid), EID join, phone+email join; UNRESOLVED sentinel for the rest; write customer_identity_xref	§6, §11	■
F12	decision_input	Assemble unified row per decision_id; Delta MERGE keyed on decision_id; populate traceability columns	§5	■
F13	Immutable snapshots	Per-decision JSON (raw records + bronze URIs + hashes) → Object-Lock bucket; content_sha256 → decision_input_snapshot Delta index	§7	■

ID	Workstream	Deliverable	SPEC ref	Launch-critical
F14	Must-pass DQ	Blocking checks (nulls/uniqueness, score ranges, enum, timestamp window, identity resolved); quarantine on fail; set dq_pass	§8	■
F15	Athena	Workgroup, _athena_results output, catalog registration; smoke queries	§3, §4	■
F16	Minimal risk mart	portfolio_monitoring_daily (core columns only — volumes, approval rate, avg scores) so risk has day-1 visibility	§9	■
F17	Orchestration	Amazon MWAA (Airflow) DAG: ingest → Silver → identity → decision_input → snapshot → DQ → mart; retries, SLAs	§3	■
F18	DDL + repo hygiene	athena/ddl/ DDLs matching SPEC §5/§9 typing rules; naming-standard lint in CI	§5, §9, §10	■

Exit criteria for Day 30 (go/no-go gate):

- An application flows end-to-end and produces a decision_input row **with** a matching Object-Lock snapshot and a verifiable content_sha256.
- Must-pass DQ demonstrably quarantines a deliberately bad row (it does not reach the mart).
- Deterministic identity resolution attaches AECB (EID) and Fraud (phone+email); AML + partial matches land as UNRESOLVED (nothing dropped).
- Athena returns the minimal risk mart; risk team can see day-1 volumes.
- Everything runs on schedule from MWAA in pre, in **me-central-1**.

3. Day 31–60 — Hardening

Theme: turn the launch slice into a production-grade, observable, recoverable, compliance-reviewed system. Mostly **LAUNCH-CRITICAL for a robust launch**; a few items are explicitly POST-LAUNCH.

ID	Workstream	Deliverable	SPEC ref	Class
H1	Probabilistic matching	Scorer (weighted Jaro-Winkler on name, exact on dob/phone/email/eid); thresholds MATCH=0.85, REVIEW=0.70; needs_manual_review flag; survivorship POSTGRES>AECB>FRAUD>AML	\$6, \$11	Launch-critical
H2	Full DQ scorecard	Add WARN tier (completeness ≥ 0.75 for $\geq 95\%$, AML PENDING $\leq 5\%$, source-freshness SLAs); write dq_scorecard_daily with 0–100 dq_score; Soda Core checks in dq/soda/	\$8	Launch-critical
H3	Alerting	SNS topics wired to DQ must-pass failures, WARN-threshold breaches, freshness-SLA misses, pipeline failures; routed to on-call	\$8	Launch-critical
H4	Portfolio mart (full)	Complete portfolio_monitoring_daily grain snapshot_date $\times$ product_code $\times$ outcome_band $\times$ risk_band — approval rate, avg fraud, AML-hit rate, PEP exposure, avg AECB, requested/approved AED, day dq_score	\$9	Launch-critical
H5	Metabase	Metabase on ECS Fargate; risk dashboards on the mart (portfolio + DQ scorecard); access controls	\$9	Launch-critical
H6	Backfill / replay	Idempotent reprocessing by partition; bookmark reset runbook; Delta MERGE re-runs proven safe; snapshot write-once respected	\$3, \$7	Launch-critical
H7	Observability	CloudWatch dashboards + metrics (per-source volumes, freshness, resolution rate, DQ pass rate, job durations); structured logs; run-book links on alarms	\$8	Launch-critical
H8	Disaster recovery	Cross-region/versioned backups (excl. Object-Lock which is inherently durable), RTO/RPO defined + tested restore of catalog + Silver	compliance	Launch-critical

ID	Workstream	Deliverable	SPEC ref	Class
H9	Security review	Least-privilege IAM audit, KMS key policy review, secrets in Secrets Manager, network review, PII-tag coverage check	§10, security	Launch-critical
H10	UAE CB audit-trail validation	Formal test: reconstruct N random past decisions from Object-Lock snapshots; prove immutability + tamper detection; document retention (7 yr) + residency	§7, §10	Launch-critical
H11	Stewardship queue (basic)	Surface needs_manual_review rows to a simple queue/Athena view for manual resolution	§6	POST-LAUNCH (basic in this tranche)
H12	Load smoke	25–50K/day synthetic burst to find early bottlenecks ahead of the full 90-day load test	§1	Launch-critical

Exit criteria for Day 60:

- Probabilistic matching lifts resolution rate to $\geq 97\%$; review-band rows are flagged and queued.
- Full DQ scorecard + SNS alerting live; a breached WARN threshold actually pages on-call.
- Risk team is using Metabase dashboards off the full portfolio mart.
- **Security review and UAE-CB audit-trail validation both signed off** (compliance gate).
- DR restore tested; backfill/replay runbook exercised once.

4. Day 61–90 — Scale & optimise

Theme: prove and cost the path to 100K/day, and lay POST-LAUNCH groundwork. Almost everything here is **POST-LAUNCH** — it is not required for the 10K/day go-live but de-risks the 12-month scale target.

ID	Workstream	Deliverable	SPEC ref	Class
S1	100K/day load test	Synthetic 10× load; measure freshness, job durations, MSK lag, cost; find + fix bottlenecks	§1, §3	POST-LAUNCH
S2	EMR migration eval	Run the identity + mart jobs on EMR / EMR-Serverless (same PySpark); compare cost + runtime vs Glue; recommend hybrid split	§3	POST-LAUNCH

ID	Workstream	Deliverable	SPEC ref	Class
S3	Cost optimisation	S3 lifecycle (cold bronze → IA/Glacier), Athena partition projection + result reuse, Glue DPU right-sizing, reserved capacity where justified	§3	POST-LAUNCH
S4	Compaction / layout	Scheduled Delta OPTIMIZE + Z-ORDER on master_customer_id / decision_id; add product_code sub-partition on hot marts; vacuum policy (retain > snapshot needs)	§3	POST-LAUNCH
S5	Advanced MDM	Full stewardship UI/queue, merge/split of golden records, SCD2 history surfacing, survivorship overrides with audit	§6	POST-LAUNCH
S6	ML groundwork	Feature tables off Silver/Gold for fraud + credit models; leakage-safe feature_as_of < label_as_of; offline eval; no scoring in the credit decision path until model-risk sign-off	§5, §8	POST-LAUNCH
S7	Post-launch roadmap	Costed 12-month roadmap: 100K/day cutover plan, EMR hybrid, ML rollout, additional products/sources	§1	POST-LAUNCH
S8	Scale hardening	MSK partition/connector-task scaling, probabilistic-scorer bucketing/salting, DQ parallelised per partition	§3, §6, §8	POST-LAUNCH

Exit criteria for Day 90:

- 100K/day sustained in load test within freshness SLA; bottlenecks documented + addressed.
- EMR-vs-Glue recommendation delivered with numbers; hybrid split decided.
- OPTIMIZE/Z-ORDER + lifecycle cut cost-per-1K-decisions measurably below the launch baseline.
- Post-launch roadmap approved by Data Eng + Risk + Product.

5. Milestone table

Milestone	Target day	Gate	Owner
M0 — Landing zone + medallion S3 + Object Lock live in dev	Day 10	Infra review	Platform/DevOps
M1 — All four ingestion paths landing to Bronze	Day 18	Data-in-Bronze check	Data Eng
M2 — Bronze→Silver + deterministic identity + decision_input	Day 24	Silver conformance	Data Eng
M3 — Immutable snapshots + must-pass DQ + minimal mart + Athena	Day 30	Go/No-Go for pre	Data Eng Lead
M4 — Probabilistic matching + full DQ scorecard + alerting	Day 45	DQ sign-off	Data Eng
M5 — Portfolio mart + Metabase + DR + backfill	Day 54	Risk UAT	Risk + Data Eng
M6 — Security + UAE-CB audit-trail review passed	Day 60	Compliance go-live gate	Compliance + Security
GO-LIVE (10K/day, Q2 2025)	Day 60–65	Exec sign-off	Data Eng Lead + Product
M7 — 100K/day load test passed	Day 78	Perf review	Data Eng + Platform
M8 — EMR eval + cost optimisation + roadmap	Day 90	Roadmap approval	Data Eng Lead

6. Ownership (RACI)

Roles: DEL = Data Eng Lead · DE = Data Engineer(s) · PLT = Platform/DevOps · RISK = Risk team ·

COMP = Compliance · SEC = Security · PM = Product.

Workstream	R	A	C	I
Landing zone / networking / IAM / KMS (F1)	PLT	DEL	SEC	RISK
Medallion S3 + Object Lock (F2, F3)	PLT	DEL	COMP	RISK
RDS + Debezium + MSK (F5, F6)	DE	DEL	PLT	—
Ingestion paths (F7–F9)	DE	DEL	PLT	—
Bronze→Silver + typing/PII (F10)	DE	DEL	COMP	RISK
Identity resolution v1 + probabilistic (F11, H1)	DE	DEL	RISK	COMP
decision_input + snapshots (F12, F13)	DE	DEL	COMP	RISK
DQ (must-pass + scorecard + alerts) (F14, H2, H3)	DE	DEL	RISK	PM
Athena + marts + Metabase (F15, F16, H4, H5)	DE	DEL	RISK	PM
Orchestration (F17)	DE	DEL	PLT	—

Workstream	R	A	C	I
Observability + DR (H7, H8)	PLT	DEL	DE	RISK
Security review (H9)	SEC	DEL	PLT	COMP
UAE-CB audit-trail validation (H10)	COMP	DEL	SEC	RISK
Load test / EMR / cost / compaction (S1–S4)	DE	DEL	PLT	PM
Advanced MDM + stewardship (S5)	DE	DEL	RISK	COMP
ML groundwork (S6)	DE	DEL	RISK	COMP
Roadmap (S7)	DEL	PM	RISK	SEC

7. Risks & mitigations

#	Risk	Likelihood	Impact	Mitigation
R1	Object Lock misconfigured → snapshots mutable/deletable (regulatory breach)	Low	Critical	Compliance-mode + default retention set at bucket creation (F3); H10 validation explicitly tries and fails to delete/alter
R2	Identity resolution drops or mis-merges customers	Med	High	UNRESOLVED sentinel guarantees no drops (F11); review band + stewardship (H1, H11); survivorship rules from §6; audit every match decision
R3	Debezium/MSK CDC lag or gaps → stale spine	Med	High	CDC-freshness SLA <15m alarmed (H3); MSK partition/task scaling (S8); replay from WAL/offset runbook (H6)
R4	AECB SFTP delays/format drift break batch ingest	Med	Med	Bookmarks make reruns safe (F7); freshness alarm <24h; schema-drift guard + quarantine on Bronze parse
R5	Glue cost/runtime doesn't scale to 100K/day	Med	High	Early load smoke at 25–50K (H12); full 100K test (S1); EMR hybrid path pre-designed (SPEC §3, arch §6)
R6	Data residency / PII exposure	Low	Critical	Region pinned to me-central-1 (F1); SSE-KMS; PII tagging (F10); security review (H9)
R7	Metabase/Athena exposes PII to risk users	Low	High	Gold exposes natural keys only, no raw PII (SPEC §10); L1/L2 columns not surfaced in marts; access controls (H5)

#	Risk	Likelihood	Impact	Mitigation
R8	DQ false-positives quarantine good rows / alert fatigue	Med	Med	Two-tier design — only must-pass blocks (S8); WARN tuned against real distributions in H2; thresholds in dq/soda/versioned
R9	Scope creep pulls POST-LAUNCH work into launch window	Med	Med	Hard launch-critical vs post-launch split in this plan; M3/M6 gates enforce it
R10	Snapshot volume/cost at 100K/day	Med	Med	Snapshots are compact JSON; S3 lifecycle + Glacier for cold, but Object-Lock retention respected (S3, S4)

8. Dependencies

External / upstream:

- AECB SFTP credentials, endpoint, and XML schema/sample files (blocks F7).
- Fraud provider REST API credentials + rate limits (blocks F8).
- AML provider webhook contract + signing secret (blocks F9).
- PostgreSQL logical-replication enablement + Debezium user grants (blocks F5, F6).
- UAE Central Bank retention + residency requirements confirmed in writing (blocks H10 sign-off).

Internal / platform:

- AWS Organizations + me-central-1 access provisioned (blocks F1 → everything).
- KMS keys + IAM roles approved by Security (blocks F1, F2, F3).
- MWAA environment provisioned (blocks F17 orchestration of everything).
- Metabase hosting decision (ECS Fargate) + risk-team accounts (blocks H5).

Sequencing: F1 → F2/F3/F4 → (F5,F6 CDC ■ F7–F9 batch/API) → F10 → F11 → F12 → F13 → F14 → F15/F16 →

F17 wires it. Hardening (H) *layers on the launch slice*; Scale (S) follows go-live.

9. Compliance (UAE Central Bank)

Compliance is a **go-live gate (M6)**, not an afterthought. The platform is designed to satisfy the UAE Central Bank on five fronts:

Requirement	How the platform meets it	SPEC / plan ref
Full audit trail of every decision	Immutable per-decision JSON snapshot capturing the resolved <code>master_customer_id</code> and verbatim raw AECB/fraud/AML/profile records with Bronze URIs + record hashes — exact reconstruction of what the engine saw	\$7 · F13 · arch \$5
7-year immutable retention	Snapshot bucket in S3 Object Lock compliance mode , default retention <code>SNAPSHOT_RETENTION_YEARS = 7</code> ; neither operators nor root can alter/delete	\$7 · F3 · R1
Tamper evidence	<code>content_sha256</code> of each snapshot stored in the queryable <code>decision_input_snapshot</code> Delta index; re-hash-and-compare proves integrity	\$7 · F13 · H10
PII handling	PII tagged in Delta column comments (PII Level 1)	2
Data residency	Entire stack pinned to AWS UAE region me-central-1 ; no cross-region replication of regulated data (DR uses in-region/versioned strategy that respects residency)	F1 · H8 · R6

Audit-trail validation (H10) is an explicit, evidenced exercise before go-live: pick N random historical decisions, reconstruct each from its Object-Lock snapshot, prove the object cannot be modified or deleted, and demonstrate that a tampered byte is caught by the `content_sha256` check. The result is documented and countersigned by Compliance and Security as the M6 gate.

Identity auditability — every match decision in `customer_identity_xref` records `match_method`, `match_confidence`, `matched_on`, `needs_manual_review`, and audit timestamps (\$6), so a regulator can also audit *how* a customer's records were fused, not just the inputs.

10. Launch-critical vs post-launch — one-glance summary

LAUNCH-CRITICAL (Q2 2025, 10K/day)	POST-LAUNCH (→ 100K/day, 12 mo)
Landing zone, medallion S3, Object Lock, catalog	100K/day load-tested scale-out
All 4 ingestion paths + CDC	EMR migration (hybrid Glue/EMR)
Bronze→Silver, PII tagging, audit block	Cost optimisation + S3 lifecycle
Deterministic identity resolution + UNRESOLVED	OPTIMIZE/Z-ORDER + partition tuning
<code>decision_input</code> + immutable snapshots	Advanced MDM + full stewardship UI
Must-pass DQ (blocking)	Full ML fraud/credit scoring in-path
Full DQ scorecard + SNS alerting	Additional products/sources
Portfolio mart + Metabase	—
Probabilistic matching + survivorship	—
Observability, DR, backfill/replay	—
Security + UAE-CB audit-trail sign-off	—

The dividing principle: **anything required for a correct, compliant, observable 10K/day launch is launch-critical; anything that only matters for 10× scale, cost, or ML sophistication is post-launch.**